EC991 Topics in Macroeconomics

Week 1 · Motivation: Household Heterogeneity

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Introduction

What is this module about?

- **Macroeconomics with heterogeneity**: moving beyond the representative agent
  - households differ in their earnings and wealth
  - firms differ in their productivity, employment and capital
- **Computational methods**: above all, how to solve these models; much of the toolkit carries over to other fields
- **Data**: how to calibrate macro models and bring them to the data

There is so much exciting work in macro. If time permits, I will also touch on **search and matching**, **automation** and **New Keynesian models**.

Macro and heterogeneity?

Table 7: Unconventional DSGE Features

Year	Heterogeneous agents	Finite horizon	Unconventional expectations / preferences	Indeterminacy	Continuous Time
1980	0		18	0	
1990	10	38	30	10	13
2000	10	10	27	0	5
2006–10	15	6	19	6	3
2016–18	29	4	25	9	9

Note: The figures are percentages of all articles in the *JME* and *JMCB*, plus the E-designated articles in the five general interest journals. The 2006–10 figures use data from 2006, 2008 and 2010; and the 2016–18 figures use data from 2016, 2017 and 2018.

Source: Glandon et al. 2023, Journal of Economic Literature

Numerical methods?

Table 8: Solution and Fitting Techniques in Theory-Centric Research

Year	All articles: numerical methods	Of DSGE: Calibration	Of DSGE: Optimization	Of optimized DSGE: Bayesian methods
1980	9	0	0	0
1990	33	0	0	0
2000	71	89	11	0
2006–10	78	73	27	15
2016–18	81	63	37	23

Note: The second column reports the share of theory-based articles that use numerical methods to solve or fit the model. The third and fourth columns report the shares of articles with quantitative DSGE models using calibration versus optimization to fit the model. The fifth column reports the share of DSGEs with optimized fit that use Bayesian methods. See section 3.1.5 for details. The calculations are based on articles in the *JME* and *JMCB*, plus the E-designated articles in the five general interest journals. The 2006–10 figures use data from 2006, 2008 and 2010; and the 2016–18 figures use data from 2016, 2017 and 2018.

Source: Glandon et al. 2023, Journal of Economic Literature

Data?

Table 10: Econometric Methods and Data Types Over Time

	Methods		Data				
Year Year	Time series	Applied micro	Micro data	Time series	Cross section	Panel	Proprietary
1980	75	25	22	89	8	3	13
1990	62	38	28	70	14	16	32
2000	58	42	28	54	8	38	30
2006–10	46	54	41	42	13	45	41
2016–18	35	65	56	34	10	56	52

Note: The figures are the shares, expressed as percentages, of econometrics-based articles articles in the *JME* and *JMCB*, plus the E-designated articles in the five general-interest journals. The method and data attributes are defined in section 3.2. The 2006–10 figures use data from 2006, 2008 and 2010; and the 2016–18 figures use data from 2016, 2017 and 2018.

Source: Glandon et al. 2023, Journal of Economic Literature

Motivation

Why should we model household heterogeneity?

1. Because that is the way the world is ×
2. Because it matters for **policy** ✓
3. Because it matters for the **aggregate macroeconomic variables** ✓

Earnings, income and wealth

- A **flow** is measured **over a period**, such as pounds per year
- A **stock** is measured **at a point in time**, such as pounds held today

Concept	Definition	Type
Earnings	Income attributed to work: wages, salaries and the labour component of self-employment income	**Flow**
Income	Earnings plus income from assets and transfers: interest, dividends, rent, pensions and benefits	**Flow**
Wealth (net worth)	The value of assets minus outstanding debts	**Stock**

Example

- **During the year:** £30,000 in earnings + £2,000 in interest and benefits = **£32,000 annual income** (before tax)
- **At the end of the year:** £200,000 in assets − £150,000 in debts = **£50,000 net wealth**

Saving links the two: income retained rather than consumed can add to the stock of wealth

Concentration and skewness of the distributions

	Earnings	Income	Wealth
Coefficient of variation	3.69	4.19	6.81
Variance of logs	1.50	0.99	4.80
Gini indexes	0.67	0.58	0.85
Location of mean	70	74	83
99–50 ratio	17.46	14.78	96.81
90–50 ratio	4.15	3.33	11.56
Mean-to-median ratio	1.96	1.85	6.49
50–30 ratio	3.21	1.64	5.50

Kuhn and Ríos-Rull 2016

Inequality: earnings, income and wealth

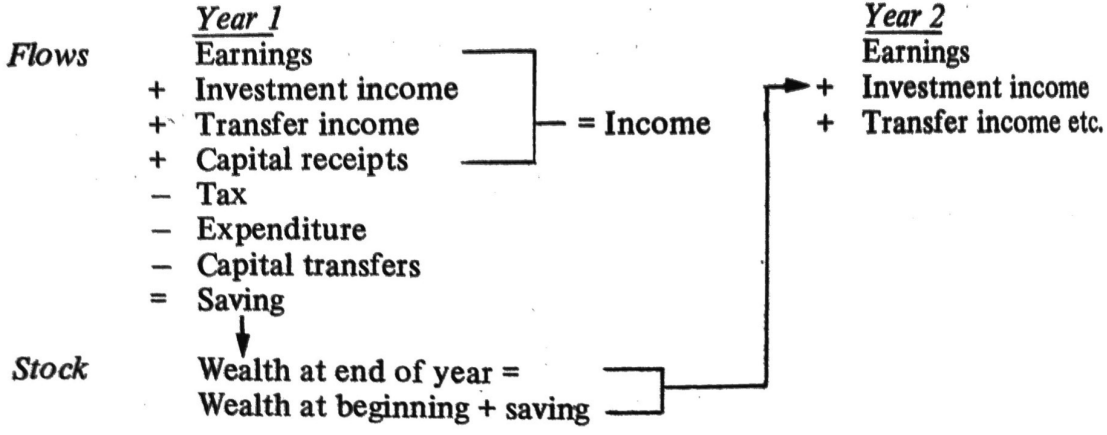
Quantiles of the 2013 earnings, income, and wealth distributions

	0	1	5	10	20	40	50	60	80	90	95	99	100
Earnings	–962.1	0	0	0	0	21.3	32.6	46.7	89.3	134.9	194.2	568.3	137,458.4
Income	–247.5	4.3	10.1	13.5	20.3	36.5	46.7	59.9	102.0	155.2	232.3	689.9	156,126.2
Wealth	–227,019.0	–78.9	–18.5	–2.0	4.3	38.2	81.4	147.6	427.8	941.0	1,871.6	7,880.4	1,324,417.5

Note: Values are 2013 thousands of dollars.

Kuhn and Ríos-Rull 2016

Budget constraint



Atkinson 1975

US: earnings inequality over time

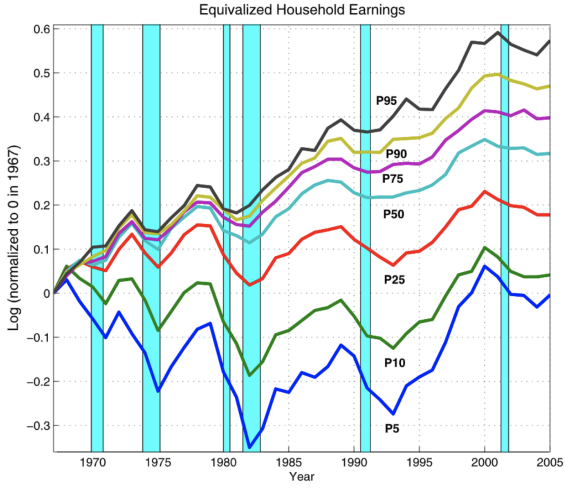
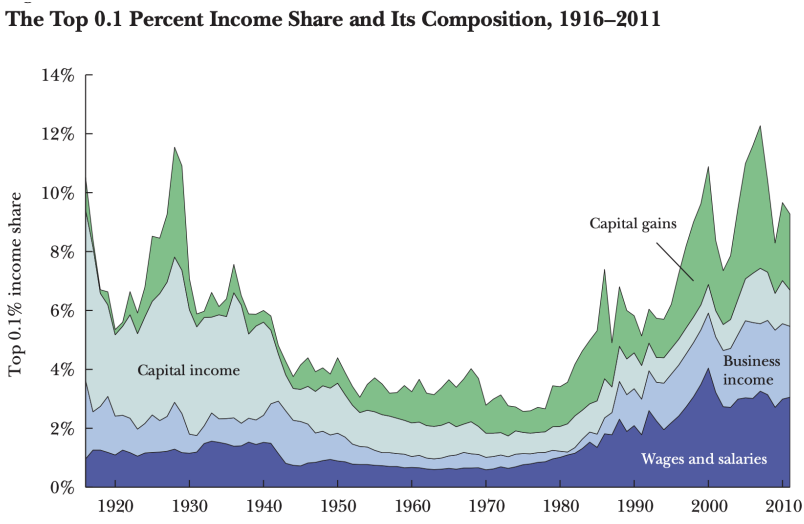


Fig. 9. Percentiles of the household earnings distribution (CPS). Shaded areas are NBER recessions.

Heathcote, Perri and Violante 2010

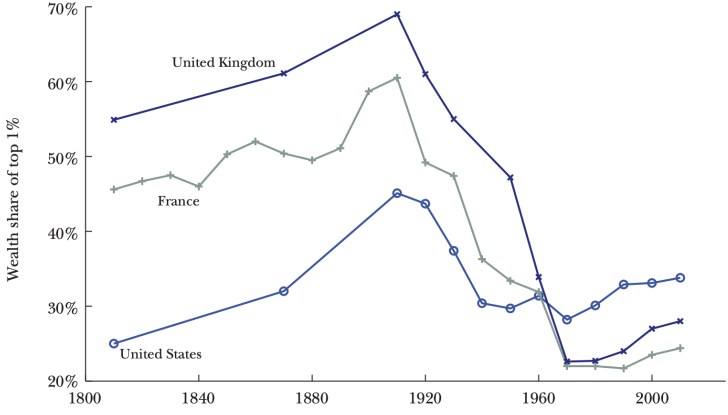
US: top 0.1%



Jones 2015

Wealth: top 1%

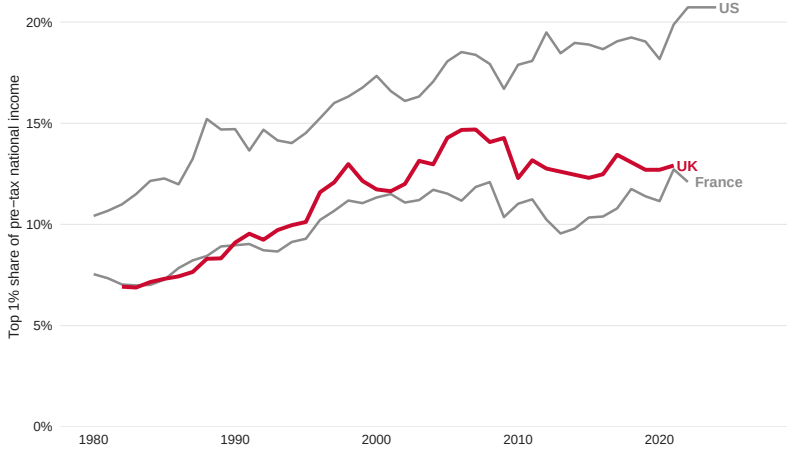
Wealth Shares of the Top 1% in Three Countries, 1800 to 2010



Source: Supplementary Table S10.1 for chapter 10 of Piketty (2014), available at: http://piketty.pse.ens.fr/capital21c.

Jones 2015

UK: top 1% income share



Notes: share of pre-tax national income received by the top 1% of adults. The UK series has a break in 1990 (tax units to individuals). The UK share roughly doubled between 1980 and 2007 and has drifted sideways since. **World Inequality Database, via Our World in Data; see also Bourquin, Brewer and Wernham 2022 (IFS Deaton Review)**

Opening the household portfolio

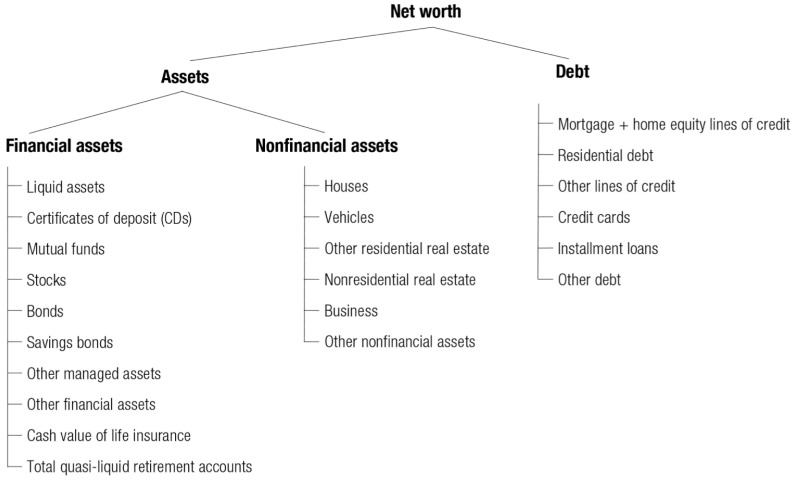
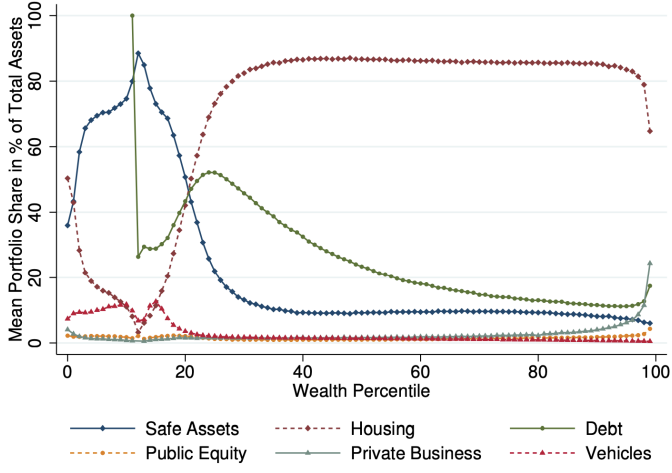


Figure 7. **SCF household portfolio.**

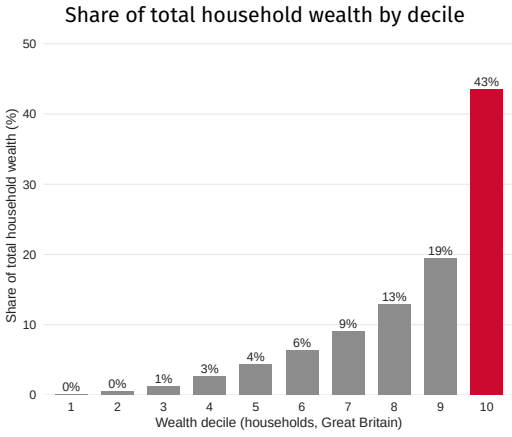
Kuhn and Ríos-Rull 2016

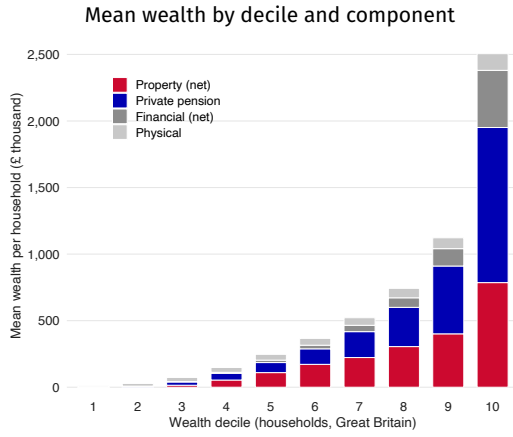
Norway: portfolio shares



Notes: 15th percentile = zero net worth. Safe assets = deposits + bonds + informal loans. Fagereng et al. 2021

UK: who holds the wealth, and in what form

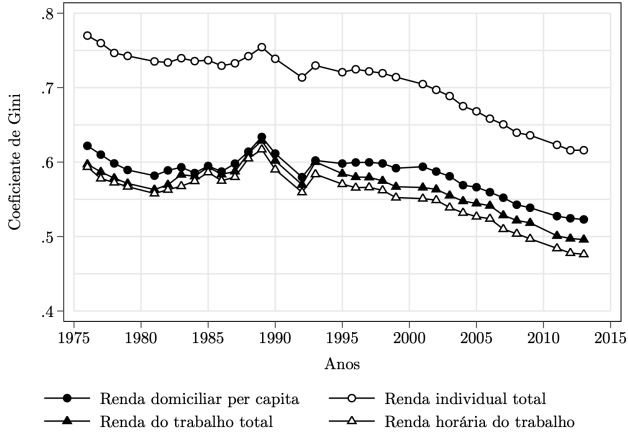




Notes: households, Great Britain, April 2018 to March 2020 (round 7). Top decile holds **43%** of measured wealth, bottom half **9%**. Financial wealth is net of unsecured debt, property net of mortgages. The survey under-covers the very top (Advani, Bangham and Leslie 2021). **ONS Wealth and Assets Survey, ad hoc release 14201**

Brazil: earnings inequality over time

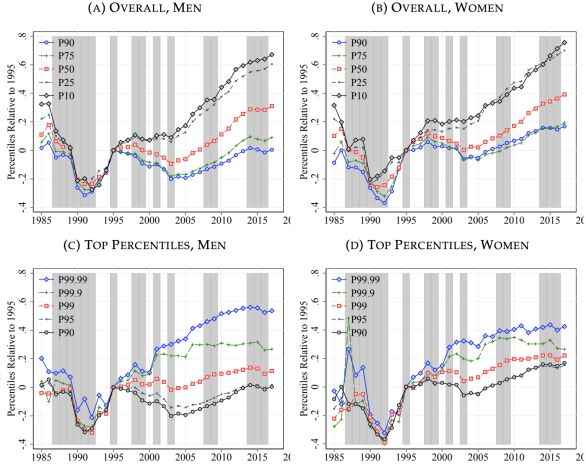
Figura 8. Coeficientes de Gini para a renda domiciliar *per capita*, renda individual total, renda total do trabalho e renda horária do trabalho principal – Brasil, 1976–2013



Souza 2016

Brazil: formal earnings inequality over time

FIGURE 2. EVOLUTION OF EARNINGS PERCENTILES, BY GENDER

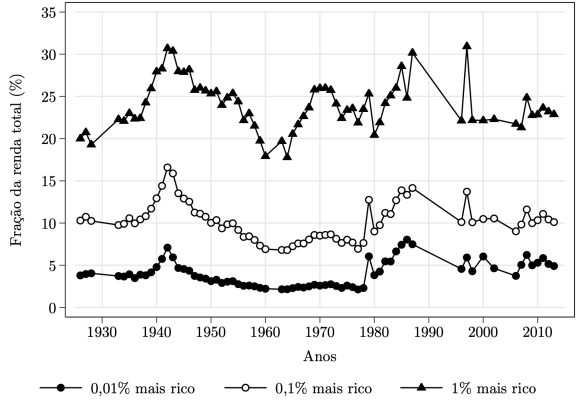


Note: Workers aged 25–55. Percentiles of the distribution of log real annual earnings, normalized to 1995. *Source:* RAIS 1985–2017.

Engbom et al. 2021

Brazil: earnings inequality over time

Figura 30. Fração da renda recebida pelos 0,01%, 0,1% e 1% mais ricos – Brasil, 1926–2013 (%)

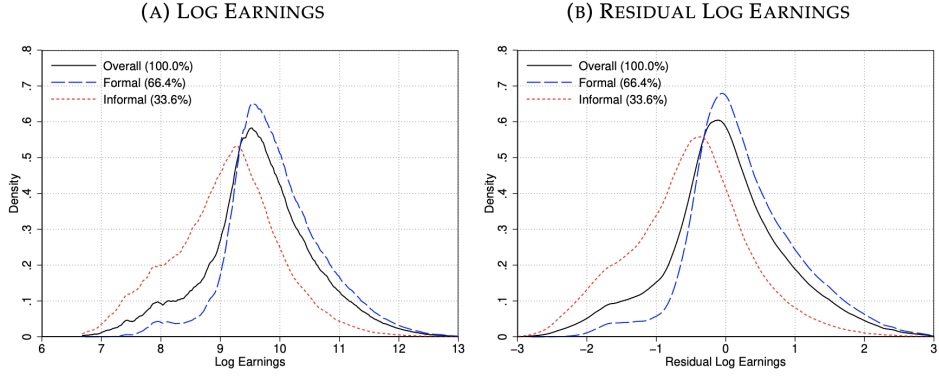


Fonte: elaboração própria a partir de tabulações de dados tributários e das Contas Nacionais; ver capítulo 4.

Souza 2016

Brazil: formal and informal earnings

FIGURE 14. DENSITIES OF (RESIDUAL) LOG EARNINGS, BY SECTOR

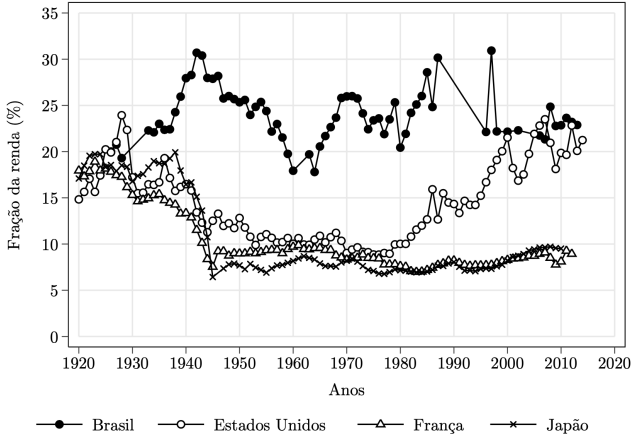


Note: Workers aged 25–55. Kernel densities of log earnings (Panel A) and residual log earnings (Panel B) in each sector. Residual log earnings are calculated controlling for age and survey wave fixed effects, separately by gender and year. Formal sector includes all employees with a work permit. Informal sector includes all employees without a work permit and the self-employed. *Source:* PME, 2002–2015.

Engbom et al. 2021

Cross-country comparison

Figura 44. Fração recebida pelo 1% mais rico no Brasil e em países desenvolvidos selecionados, 1920–2013



Souza 2016

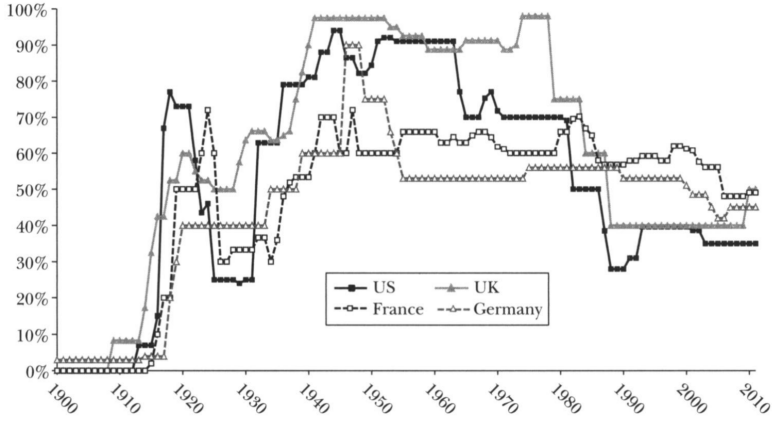
Policy

Should we do something?

- Redistributive policies have **strong welfare effects** for the poor
- What are the trade-offs? Do transfers induce **behavioural responses**?
  - Do people stop working? Do they adjust their savings?
- How should we finance them?
  - Consumption tax? Income tax? Wealth tax? What is the optimal level of **progressivity**?

Policy: top taxes

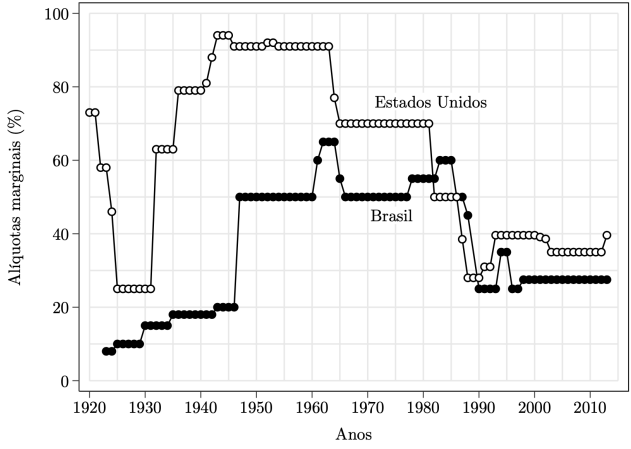
Figure 3
Top Marginal Income Tax Rates, 1900–2011



Alvaredo et al. 2013

Policy: top taxes

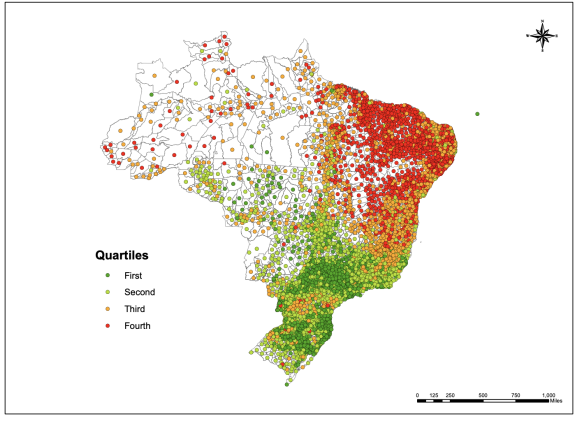
Figura 16. Alíquotas marginais máximas do IRPF no Brasil e nos Estados Unidos, 1920–2013



Souza 2016

Policy: cash transfers

(a) Number of PBF beneficiaries as a share of the population in 2008



Gerard, Naritomi and Silva 2021

Does redistribution change total consumption?

Experiment. Transfer **£100** from a rich household to a poor household

- Hold prices and their combined resources fixed
- **MPC**: the change in consumption following an extra £1 of available resources

	Common MPCs
Poor household's MPC	0.4
Rich household's MPC	0.4
Poor household's consumption change	+£40
Rich household's consumption change	−£40
Change in total consumption	**£0**

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	Common MPCs	**Different MPCs**
Poor household's MPC	0.4	0.8
Rich household's MPC	0.4	0.2
Poor household's consumption change	+£40	+£80
Rich household's consumption change	−£40	−£20
Change in total consumption	**£0**	**+£60**

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Change in total consumption	**£0**	**+£60**

$$\Delta C = (m_{\text{poor}} - m_{\text{rich}})\, T$$

- Common MPCs: gains and losses in spending **cancel**
- Different MPCs: **who receives** the resources matters
- Welfare can change in either case

When can we aggregate households?

Condition. At given prices, demand for each good has the **same slope in resources** across households

Gorman's preference restriction.

$$v_h(p, w_h) \;=\; \alpha_h(p) + b(p)\, w_h, \qquad b(p) \text{ common across households}$$

p: prices; w_h: household resources; v_h: utility after choosing the best affordable bundle

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Demand, by Roy's identity.

$$x_{ih} \;=\; -\frac{\partial v_h/\partial p_i}{\partial v_h/\partial w_h} \;=\; \underbrace{-\frac{\partial \alpha_h/\partial p_i}{b}}_{\text{household-specific intercept}} + \underbrace{\left(-\frac{\partial b/\partial p_i}{b}\right)}_{\text{common slope}} w_h$$

- Households can differ in their needs and consumption levels
- An extra pound changes each good's demand by the **same amount**
- Aggregate demand depends on **total resources**, not their distribution

▸ Worked example

Gorman 1953. Dynamic versions: Rubinstein 1974; Chatterjee 1994; Caselli and Ventura 2000

Macro aggregates

- Does inequality matter for the aggregate variables?
- Benchmark: representative agent, or homothetic preferences with complete markets (Gorman)
- The benchmark predicts **no direct consumption effect** from redistribution
- Does the evidence support **common MPCs**?

Heterogeneity in the marginal propensity to consume

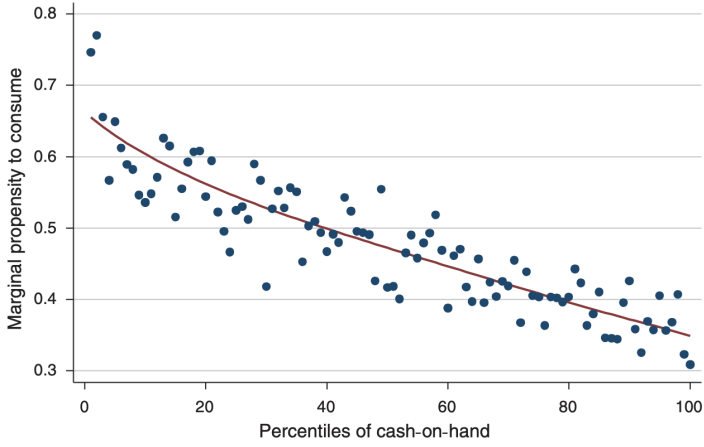


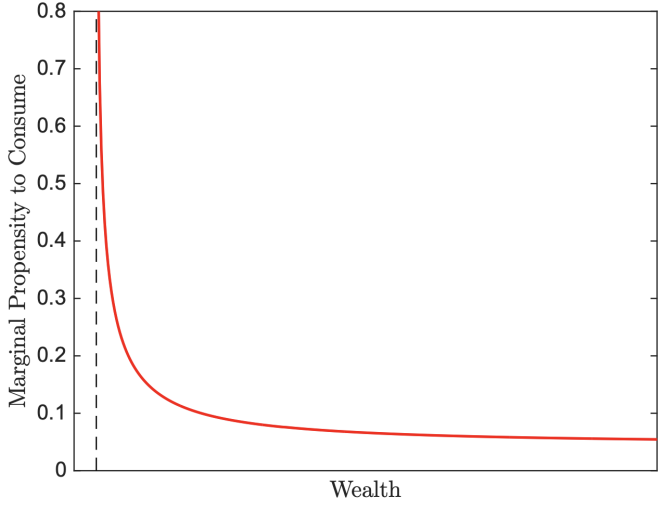
FIGURE 2. AVERAGE MPC BY CASH-ON-HAND PERCENTILES

Jappelli and Pistaferri 2014

Macro aggregates

- Does inequality matter for the aggregate variables?
- **Early heterogeneous-agent models**: small effects on consumption and saving.
  "Quasi-aggregation" $\Rightarrow$ rich, middle class and poor behave the same, scaled by income
  - only the very poor behave differently, and they are too small to move aggregates
- Still: policies can have **large welfare effects** (social security, progressive taxation, …)
- Macro (policies) affect inequality, but inequality has **little effect on macro**

Heterogeneity in MPC



Macro aggregates

- Does inequality matter for the aggregate variables?
- **Modern models**: inequality matters
  - households hold assets of different **liquidity** (housing, bonds, stocks), with different MPCs
  - separating liquid from illiquid assets: **high MPCs** even among households with substantial total wealth
  - so the **composition of wealth** matters for predicting consumption responses
- Strong feedback between the distribution and the aggregates: **macro** $\Leftrightarrow$ **inequality**
- Policies move the distribution, and the distribution **reinforces or attenuates** the effect of the policy on the aggregates

Macro $\Leftrightarrow$ inequality: a feedback loop

Inequality here is the distribution of income, liquid assets, illiquid assets and debts

Macro $\Rightarrow$ inequality. The same shock hits households differently

- Recession: job losses concentrate among lower earners, so earnings inequality rises
- Interest-rate change: borrowers, savers and asset owners are affected differently

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Inequality $\Rightarrow$ macro. Two economies lose the same £1,000 of aggregate income

Where the loss falls	MPC of affected households	Initial consumption cut
Households with little liquid savings	0.8	**£800**
Households with ample liquid savings	0.2	**£200**

Illustrative numbers

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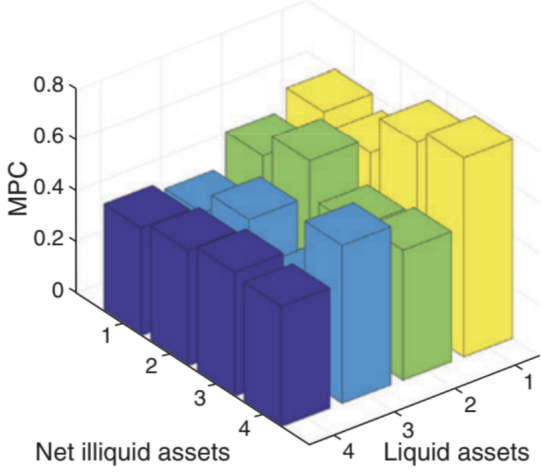
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Illustrative numbers

Put the arrows together.

- Income falls $\Rightarrow$ households with little liquid savings cut spending $\Rightarrow$ firms' sales and jobs fall $\Rightarrow$ incomes fall again
- The distribution can **amplify or cushion** the shock; how much depends on the model and environment

Heterogeneity in the marginal propensity to consume



Fagereng, Holm and Natvik 2021

UK: the MPC depends on the sign of the shock and the balance sheet

Mean MPC out of an unexpected, temporary income change: **0.14** after a gain, **0.64** after a loss

Figure 4: Mean MPC by total debt to income ratio [a]

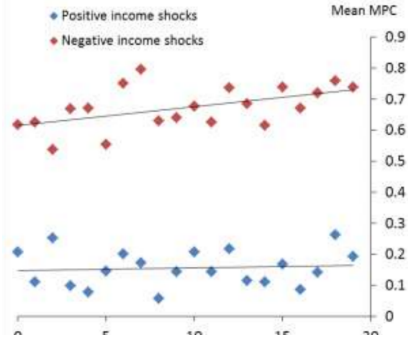
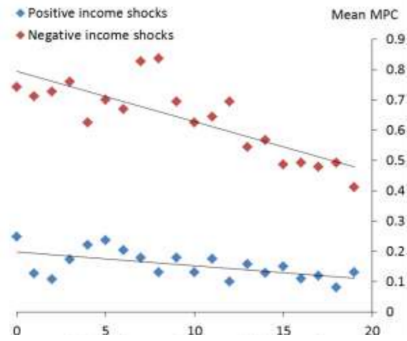
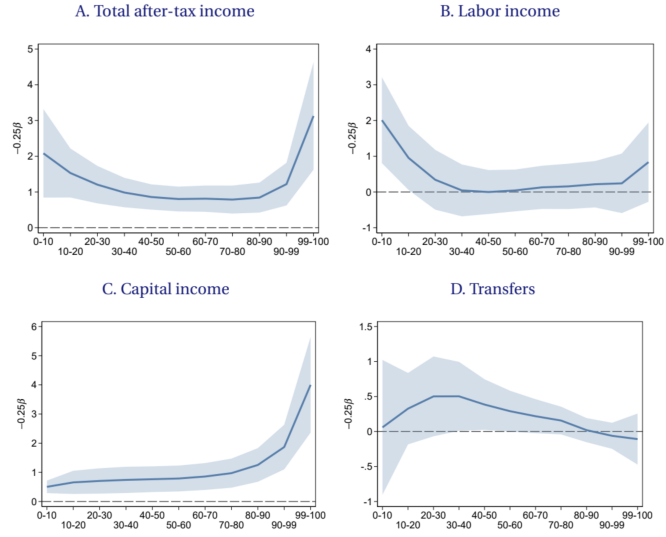


Figure 5: Mean MPC by net liquid assets to income ratio [a]



Notes: British households, Bank of England/NMG survey 2011 to 2014. Each dot is a 5% group of the debt-to-income (left) or liquid-assets-to-income (right) distribution. The MPC out of a loss is highest for indebted and illiquid households. **Bunn, Le Roux, Reinold and Surico 2018, JME**

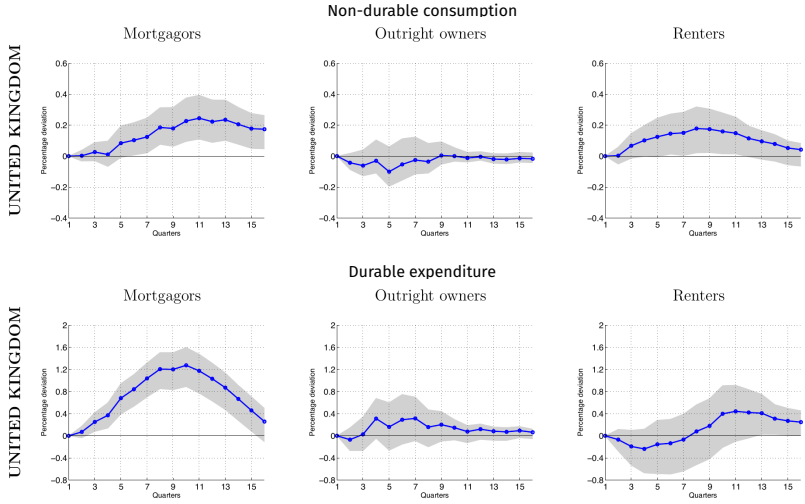
Heterogeneity in the effects of monetary policy



Amberg et al. 2022, AER: Insights; see also Almgren et al. 2022, AEJ: Macro

UK: monetary policy works through mortgagors

Response to a 25 basis point unanticipated rate cut, by housing tenure



Notes: UK Living Costs and Food Survey, 1975 to 2007; 90% bootstrapped bands. Outright owners do not respond; mortgagors do, most of all on durables. The gap is explained by liquidity: 40 to 50% of mortgagors hold net liquid wealth below half a month of income. **Cloyne, Ferreira and Surico 2020, REStud**

How we got here

- **Methodological and computational advances**: solving these models is hard; better computers help
- **Better data**: the distribution matters, and high-quality microdata uncover new facts; the credibility revolution is ongoing in applied micro
- **Policy debate**: inequality is high and rising in much of the world; growing interest from policymakers and economists

History of economic thought

A little of the history of economic thought behind this part of macro:

- Blog post by Béatrice Cherrier: here
- Paper by Béatrice Cherrier, Pedro Duarte and Aurélien Saïdi: here
- Lecture notes by Ben Moll: here

Worked example: different needs, common marginal responses

Stone–Geary preferences. Food f and clothing c, both priced at £1

$$u_h(f,c) = \sqrt{(f-\bar{f}_h)(c-\bar{c}_h)}, \qquad w_h > \bar{f}_h + \bar{c}_h$$

Demands.

$$f_h = \bar{f}_h + \tfrac{1}{2}\left(w_h - \bar{f}_h - \bar{c}_h\right), \qquad c_h = \bar{c}_h + \tfrac{1}{2}\left(w_h - \bar{f}_h - \bar{c}_h\right)$$

- Meet **basic needs** $(\bar{f}_h, \bar{c}_h)$ first
- Split the remaining budget equally
- Everyone spends another **50p** on each good per additional £1

Link to Gorman

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Worked example: the Gorman form

Indirect utility at general prices (p_f, p_c)

$$v_h(p, w_h) = \underbrace{-\frac{p_f \bar{f}_h + p_c \bar{c}_h}{2\sqrt{p_f p_c}}}_{\alpha_h(p)} + \underbrace{\frac{1}{2\sqrt{p_f p_c}}}_{b(p)} w_h$$

- Basic needs enter only the intercept $\alpha_h(p)$
- $b(p)$ is the same for every household
- **Different average spending shares** can coexist with **identical marginal responses**

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